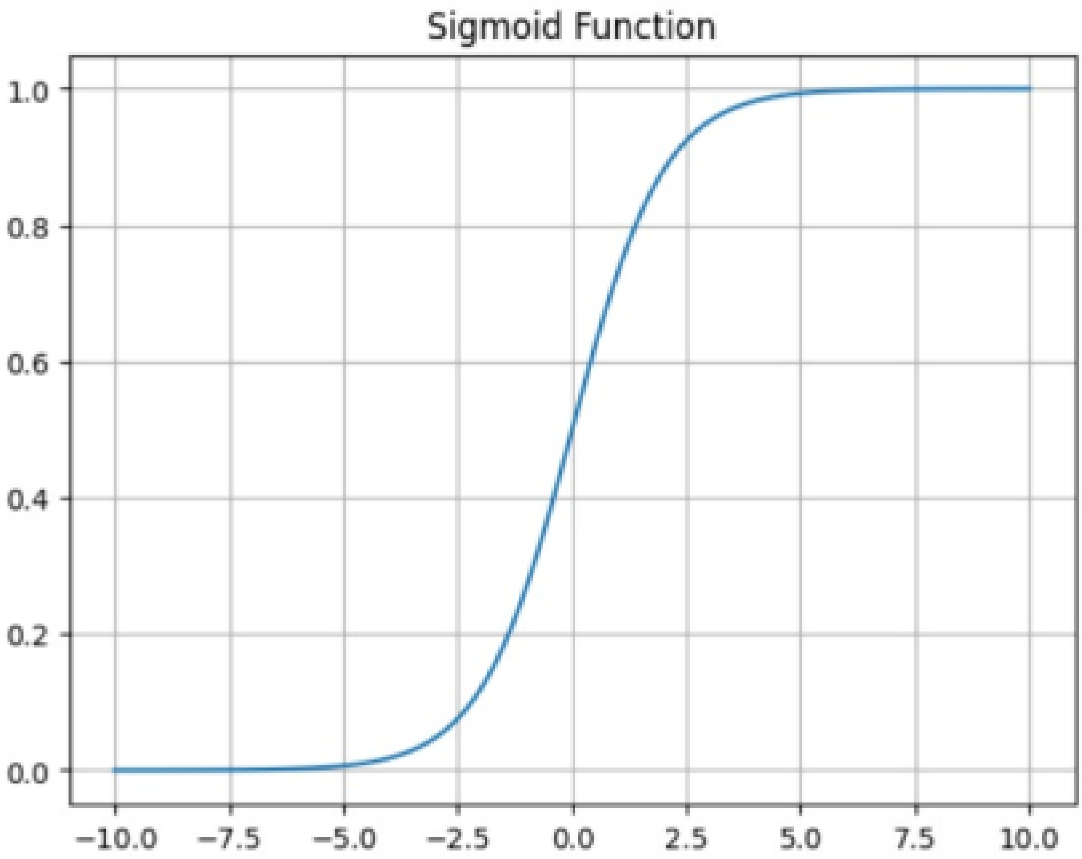
EXP:6

```
from sklearn.datasets import load_wine
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
X,y=load_wine(return_X_y=True)
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=42)
model=KNeighborsClassifier(n_neighbors=3).fit(X_train,y_train)
print("Accuracy:",accuracy_score(y_test,model.predict(X_test)))
```

```
Accuracy: 0.7407407407407407
```

EXP:7

```
import numpy as np
import matplotlib.pyplot as plt
x = np.linspace(-10,10,100)
y = 1/(1+np.exp(-x))
plt.plot(x,y)
plt.title("Sigmoid Function")
plt.grid()
plt.show()
```



EXP:8

```python
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
X,y=load_iris(return_X_y=True)
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=42)
model=KNeighborsClassifier(n_neighbors=3).fit(X_train,y_train)
print("Accuracy:",accuracy_score(y_test,model.predict(X_test)))
```

```
Accuracy: 1.0
```

EXP:9

```python
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score
X,y=load_iris(return_X_y=True)
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=42)
model=GaussianNB().fit(X_train,y_train)
print("Accuracy:",accuracy_score(y_test,model.predict(X_test)))
```

```
Accuracy: 0.9777777777777777
```

EXP:10

```python
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
X,y=load_iris(return_X_y=True)
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=42)
model=LogisticRegression(max_iter=200).fit(X_train,y_train)
print("Accuracy:",accuracy_score(y_test,model.predict(X_test)))
```

```
Accuracy: 1.0
```